



Lecture 3: Stability

Chen, Jiahao
SIST 1D#206



Poles and Zeros

- Pole, zero, and gain defined by Douglas:

$$G(s) = \frac{s + 5}{(s + 1)(s + 3)}$$

Poles are values of s that cause $G(s) \rightarrow \infty$

$$s = -1 \quad \text{and} \quad s = -3$$

Zeros are values of s that cause $G(s) \rightarrow 0$

$$s = -5$$

Gain is the value of $G(s)$ under steady state conditions

$$G(0) = \frac{0 + 5}{(0 + 1)(0 + 3)} = \frac{5}{3}$$

Poles and Zeros

- ❑ Pole due to input -> forced response
- ❑ Pole due to transfer function -> natural response
- ❑ Zero and pole both contribute to the amplitudes of the response
- ❑ Recall the solution of the state space model:

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau$$

$$= \underbrace{\Phi(t) \mathbf{x}(0)}_{\text{zero-input response (natural response)}} + \underbrace{\int_0^t \Phi(t-\tau) \mathbf{B} \mathbf{u}(\tau) d\tau}_{\text{Zero-state response (forced response) as convolution integral}}$$

zero-input response
(natural response)

Zero-state response
(forced response)
as convolution integral

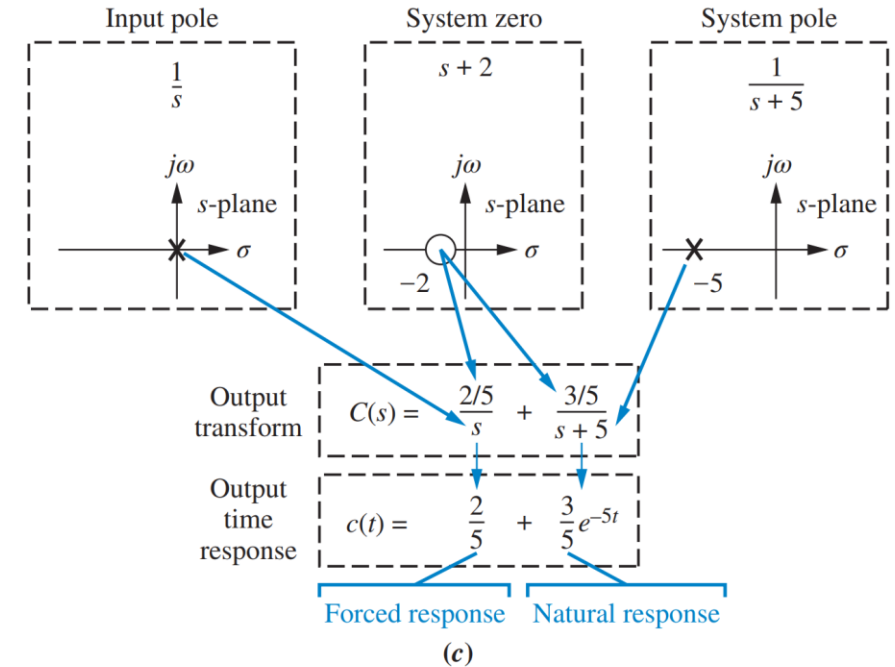
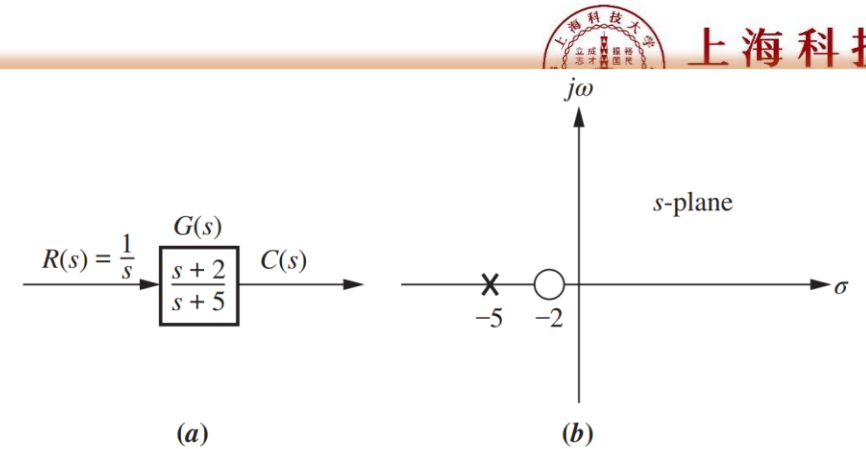


FIGURE 4.1 a. System showing input and output; b. pole-zero plot of the system; c. evolution of a system response. Follow blue arrows to see the evolution of the response component generated by the pole or zero.

Evaluating Response Using Poles

PROBLEM: Given the system of Figure 4.3, write the output, $c(t)$, in general terms. Specify the forced and natural parts of the solution.

SOLUTION: By inspection, each system pole generates an exponential as part of the natural response. The input's pole generates the forced response. Thus,

$$C(s) \equiv \underbrace{\frac{K_1}{s}}_{\text{Forced response}} + \underbrace{\frac{K_2}{s+2} + \frac{K_3}{s+4} + \frac{K_4}{s+5}}_{\text{Natural response}} \quad (4.3)$$

Taking the inverse Laplace transform, we get

$$c(t) \equiv \underbrace{K_1}_{\text{Forced response}} + \underbrace{K_2 e^{-2t} + K_3 e^{-4t} + K_4 e^{-5t}}_{\text{Natural response}} \quad (4.4)$$

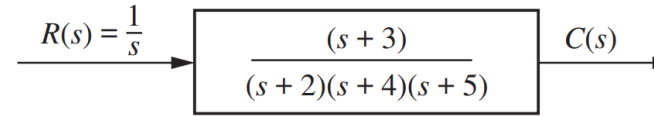


FIGURE 4.3 System for Example 4.1

PROBLEM: A system has a transfer function, $G(s) = \frac{10(s+4)(s+6)}{(s+1)(s+7)(s+8)(s+10)}$.

Write, by inspection, the output, $c(t)$, in general terms if the input is a unit step.

ANSWER: $c(t) \equiv A + Be^{-t} + Ce^{-7t} + De^{-8t} + Ee^{-10t}$



Typical Systems (excited with test signal)

- Step response of the first order system (with no zero)
 - Forced and natural response:

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$

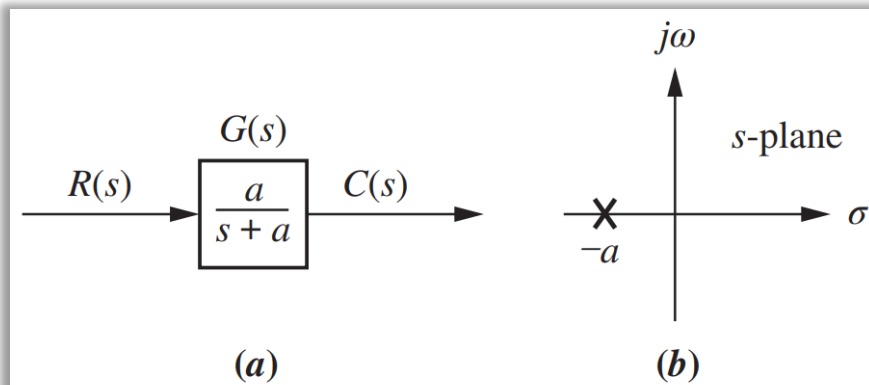


FIGURE 4.4 a. First-order system; b. pole plot

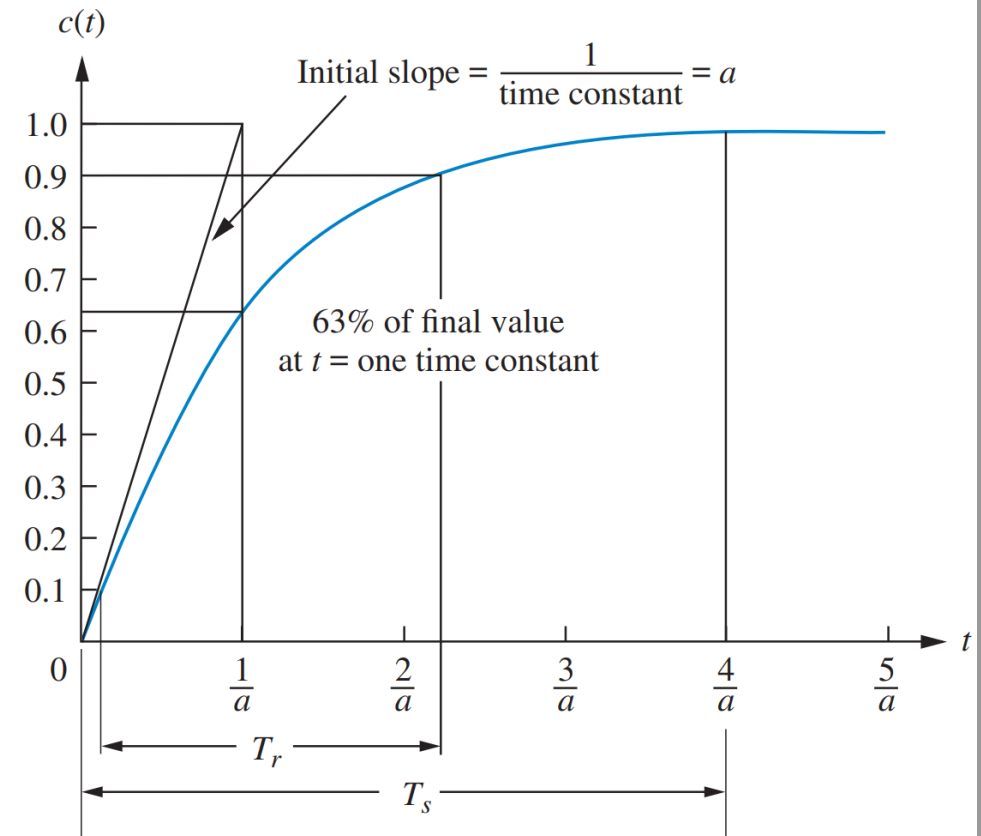


FIGURE 4.5 First-order system response to a unit step

Second order system

□ Step response of the second order system (with no zero) depends on pole locations on the pzmap:

- How about right plane pole?

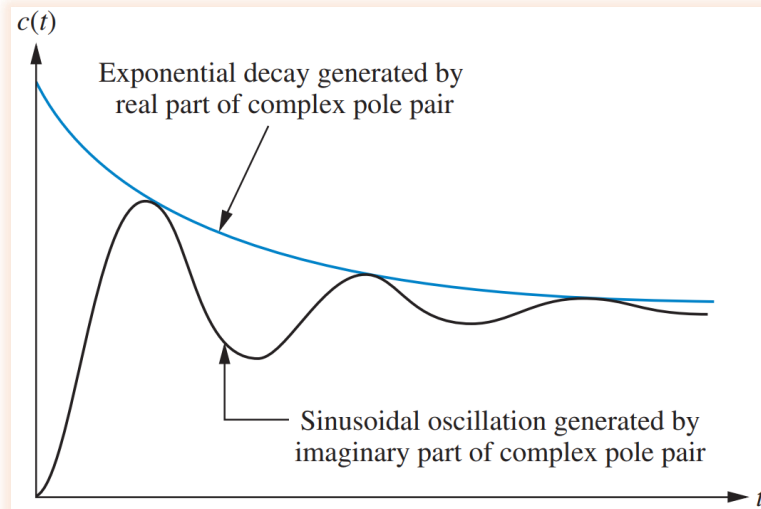


FIGURE 4.8 Second-order step response components generated by complex poles

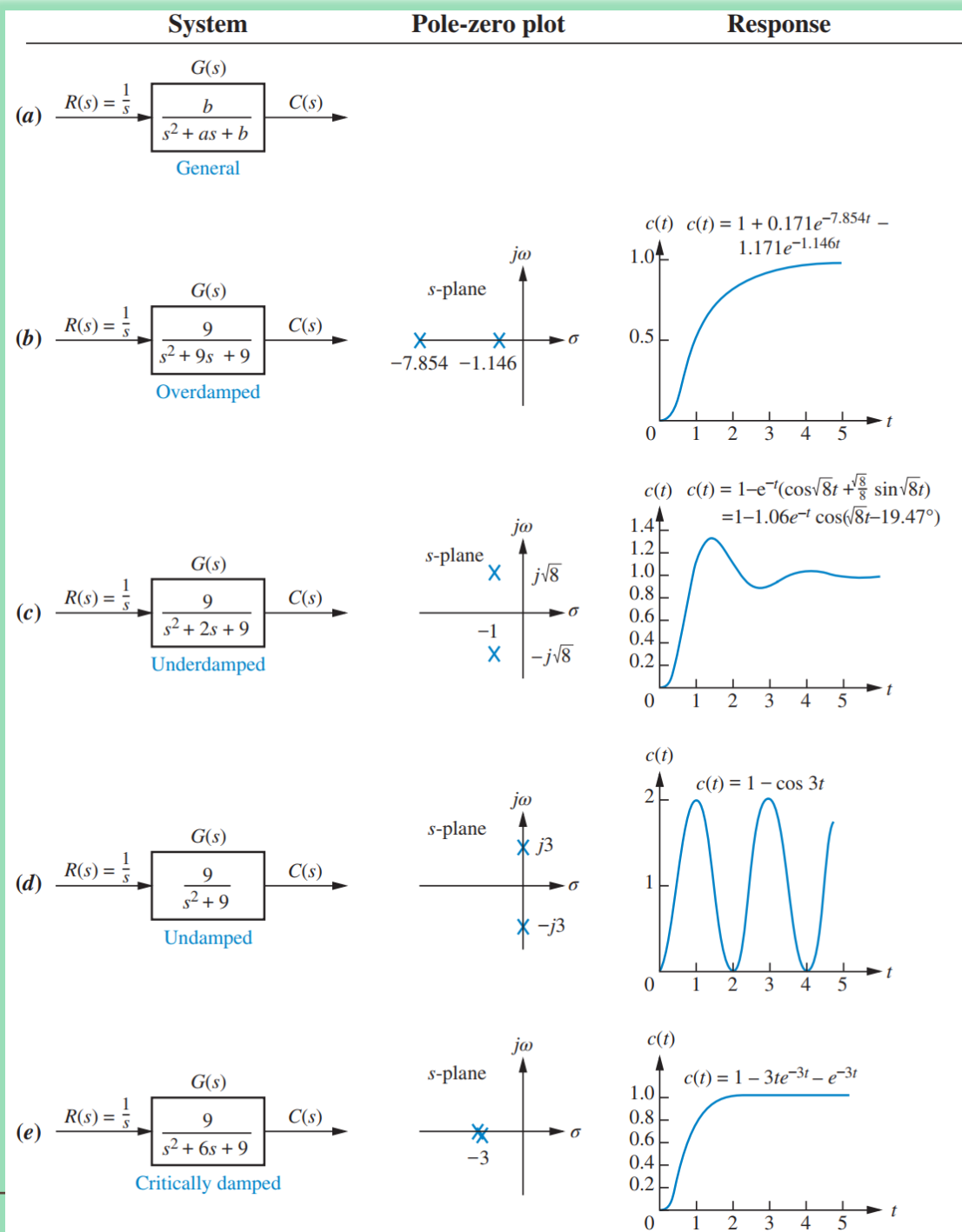


FIGURE 4.7 Second-order systems, pole plots, and step responses

- ❑ Natural frequency ω_n defines the time scale of the time response plot
 - In s-plane, it defines the length of the pole location vector
- ❑ Damping ratio ζ defines the shape of the time response plot
 - In s-plane, it defines the angle of the pole location vector

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Finding ζ and ω_n For a Second-Order System

PROBLEM: Given the transfer function of Eq. (4.23), find ζ and ω_n .

$$G(s) = \frac{36}{s^2 + 4.2s + 36} \quad (4.23)$$

SOLUTION: Comparing Eq. (4.23) to (4.22), $\omega_n^2 = 36$, from which $\omega_n = 6$. Also, $2\zeta\omega_n = 4.2$. Substituting the value of ω_n , $\zeta = 0.35$.

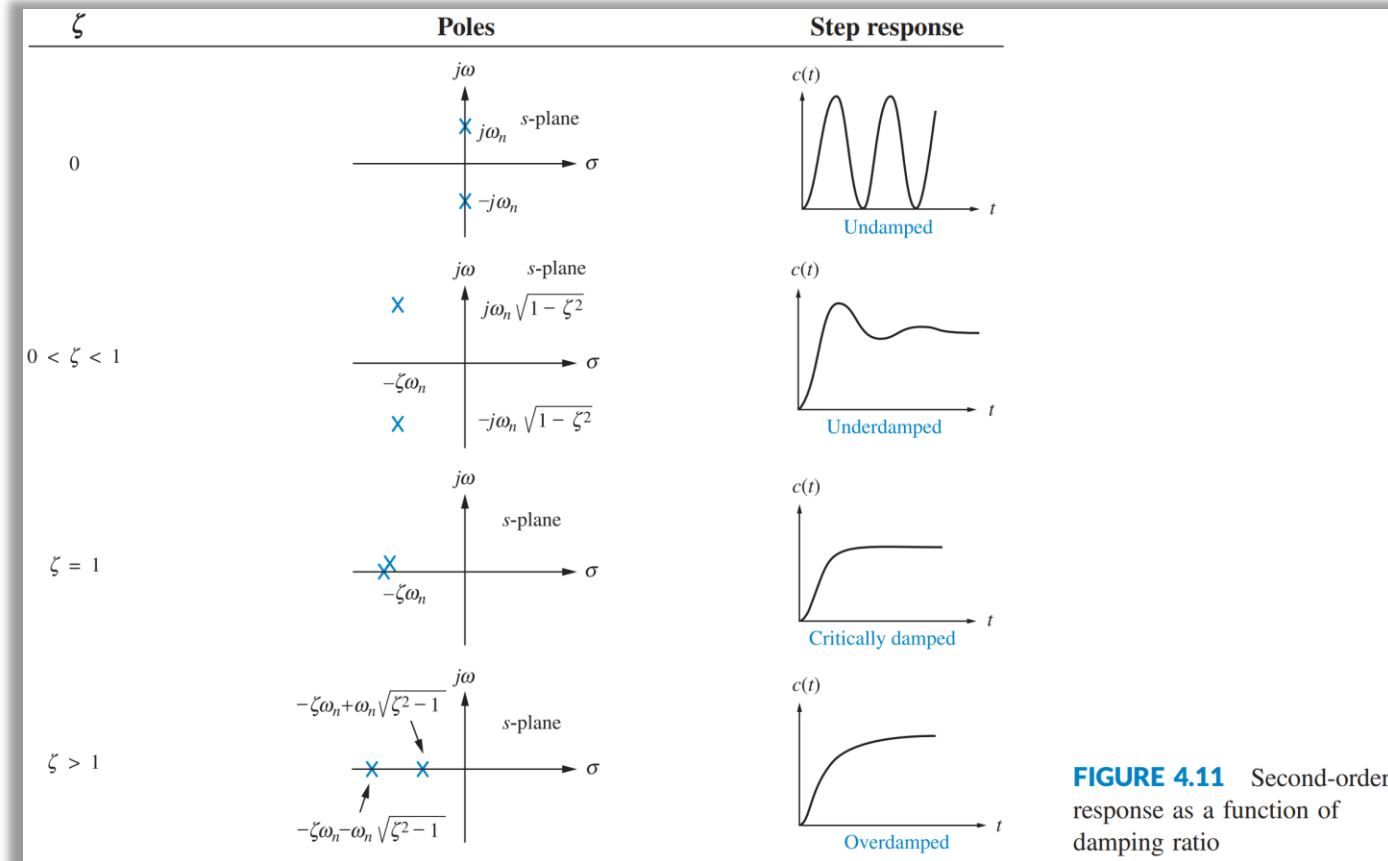


FIGURE 4.11 Second-order response as a function of damping ratio

- Time domain performance metrics are defined in terms of the underdamped response of a second order system with $\zeta \in (0,1)$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$\begin{aligned} c(t) &= 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1-\zeta^2} t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} t \right) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t - \phi) \end{aligned}$$

where $\phi = \tan^{-1}(\zeta/\sqrt{1-\zeta^2})$.

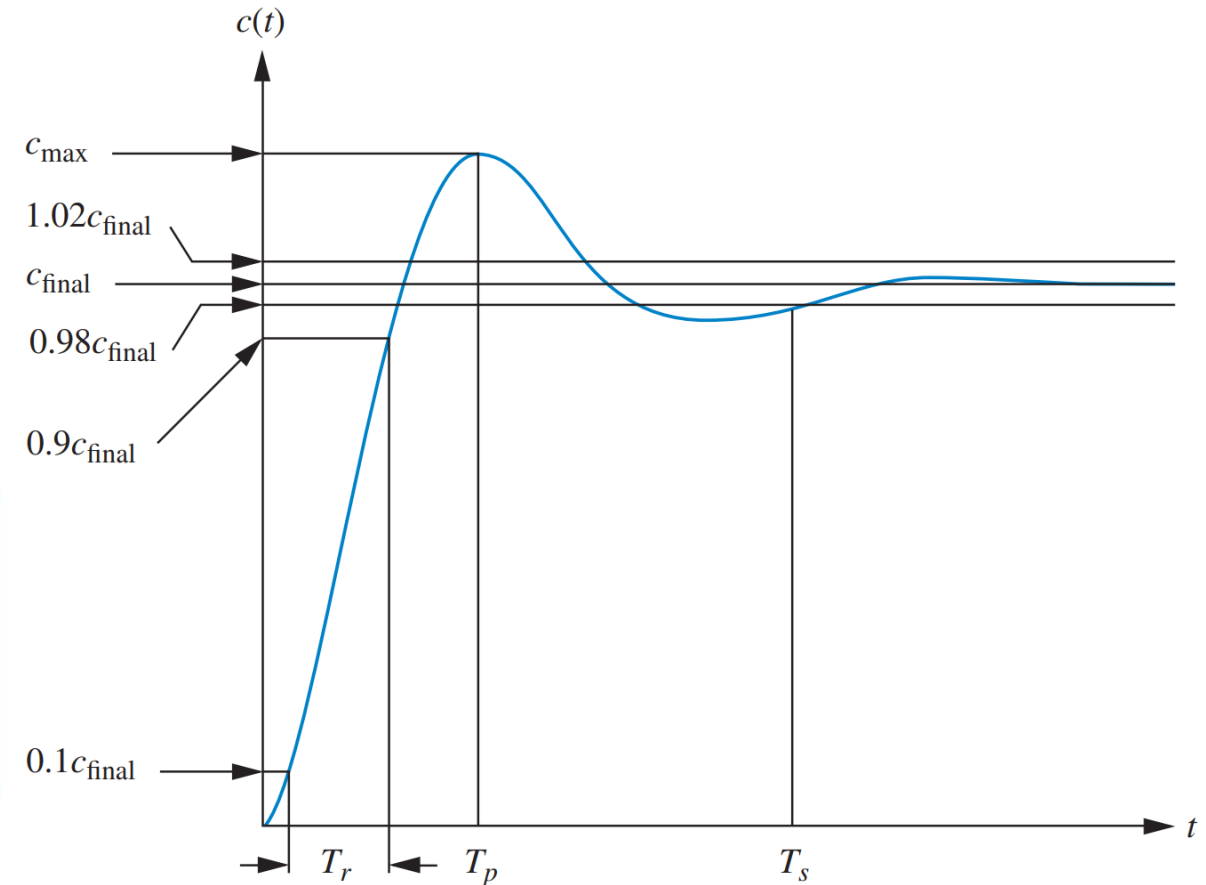


FIGURE 4.14 Second-order underdamped response specifications

Finding T_p , %OS, T_s , and T_r from a Transfer Function

PROBLEM: Given the transfer function

$$G(s) = \frac{100}{s^2 + 15s + 100} \quad (4.43)$$

find T_p , %OS, T_s , and T_r .

SOLUTION: ω_n and ζ are calculated as 10 and 0.75, respectively. Now substitute ζ and ω_n into Eqs. (4.34), (4.38), and (4.42) and find, respectively, that $T_p = 0.475$ second, %OS = 2.838, and $T_s = 0.533$ second. Using the table in Figure 4.16, the normalized rise time is approximately 2.3 seconds. Dividing by ω_n yields $T_r = 0.23$ second. This problem demonstrates that we can find T_p , %OS, T_s , and T_r without the tedious task of taking an inverse Laplace transform, plotting the output response, and taking measurements from the plot.

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (4.34)$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \quad (4.38)$$

$$T_s = \frac{4}{\zeta\omega_n} \quad (4.42)$$

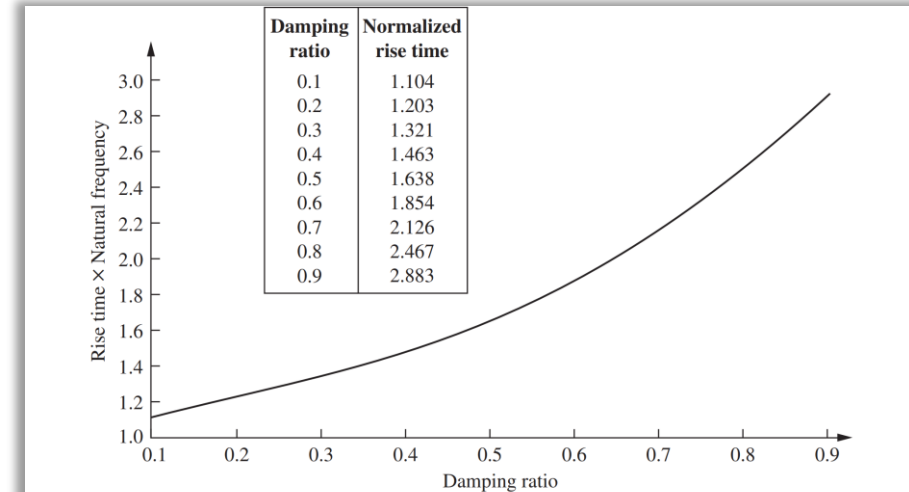


FIGURE 4.16 Normalized rise time versus damping ratio for a second-order underdamped response

Time domain performance metrics in s-domain

□ Peak time and settling time are related to :

- damped frequency of oscillation, ω_d
- and exponential damping frequency σ_d

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\zeta \omega_n} = \frac{\pi}{\sigma_d}$$

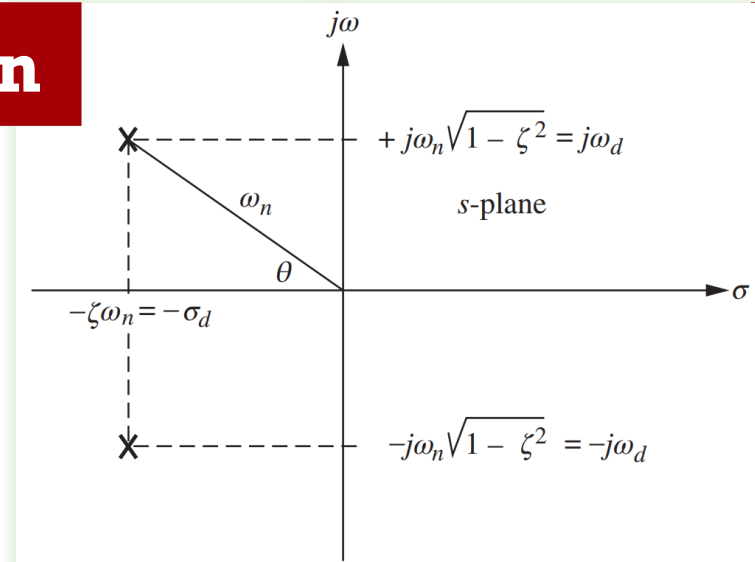


FIGURE 4.17 Pole plot for an underdamped second-order system

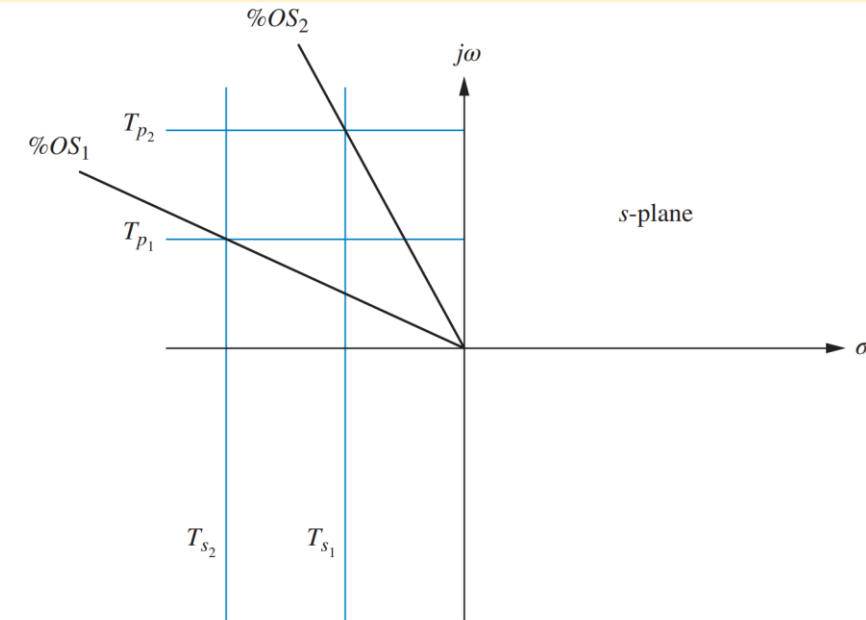


FIGURE 4.18 Lines of constant peak time, T_p , settling time, T_s , and percent overshoot, $\%OS$. Note: $T_{s2} < T_{s1}$; $T_{p2} < T_{p1}$; $\%OS_1 < \%OS_2$.

□ Example responses

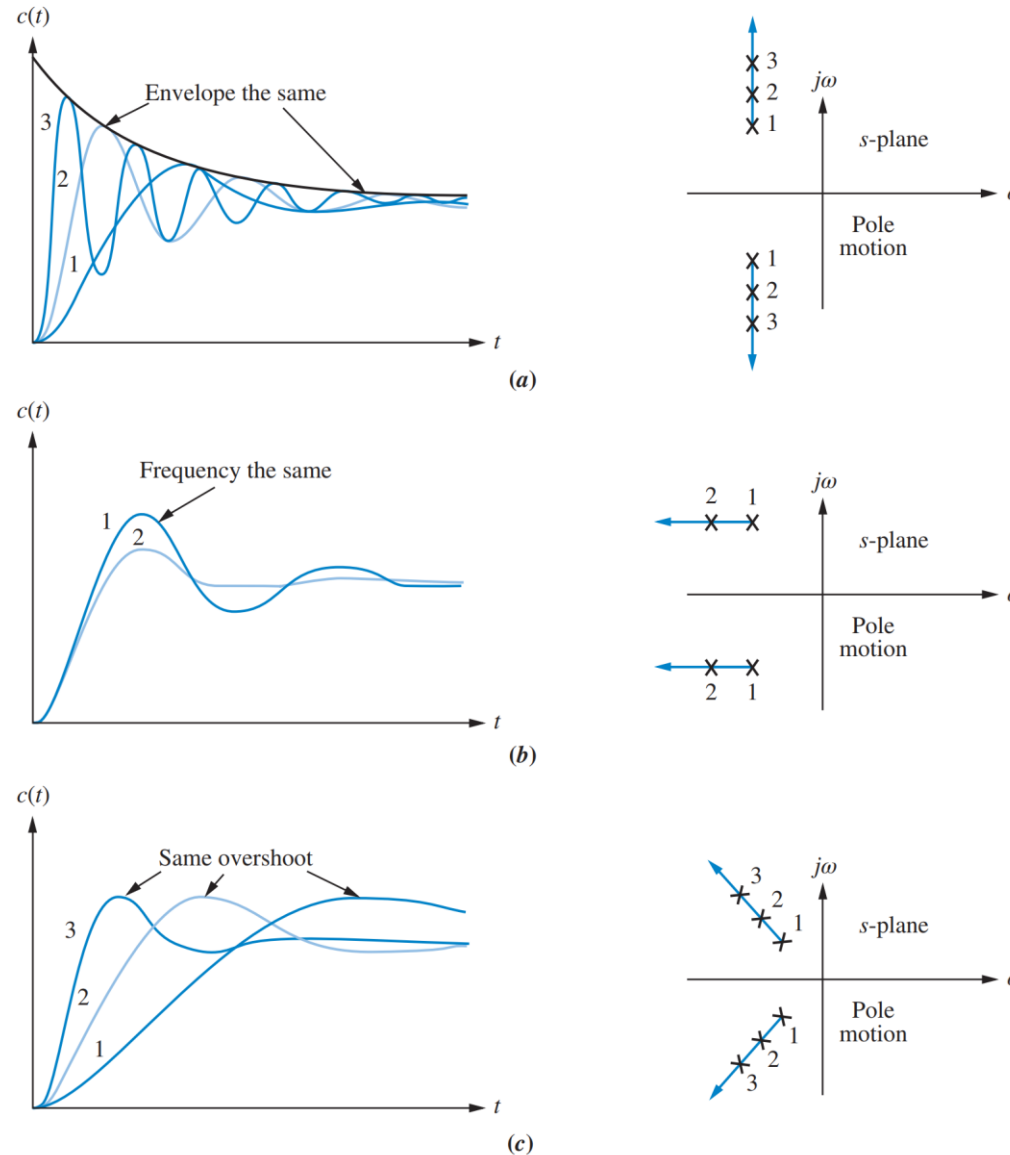


FIGURE 4.19 Step responses of second-order underdamped systems as poles move: **a.** with constant real part; **b.** with constant imaginary part; **c.** with constant damping ratio

Finding T_p , %OS, and T_s from Pole Location

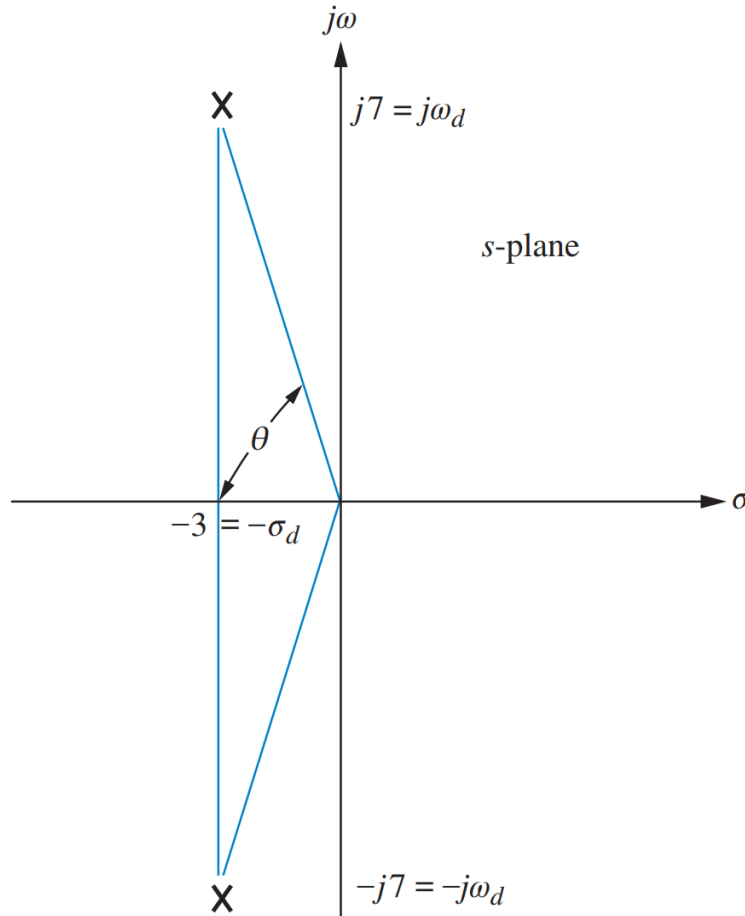


FIGURE 4.20 Pole plot for Example 4.6

PROBLEM: Given the pole plot shown in Figure 4.20, find ζ , ω_n , T_p , %OS, and T_s .

SOLUTION: The damping ratio is given by $\zeta = \cos \theta = \cos[\arctan(7/3)] = 0.394$. The natural frequency, ω_n , is the radial distance from the origin to the pole, or $\omega_n = \sqrt{7^2 + 3^2} = 7.616$. The peak time is

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{7} = 0.449 \text{ second} \quad (4.46)$$

The percent overshoot is

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 26\% \quad (4.47)$$

The approximate settling time is

$$T_s = \frac{4}{\sigma_d} = \frac{4}{3} = 1.333 \text{ seconds} \quad (4.48)$$

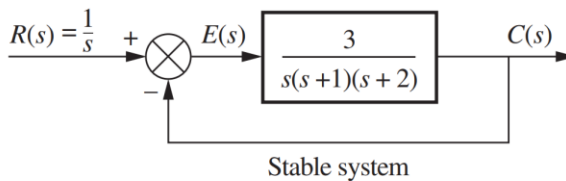


Stability **(in terms of total response)**

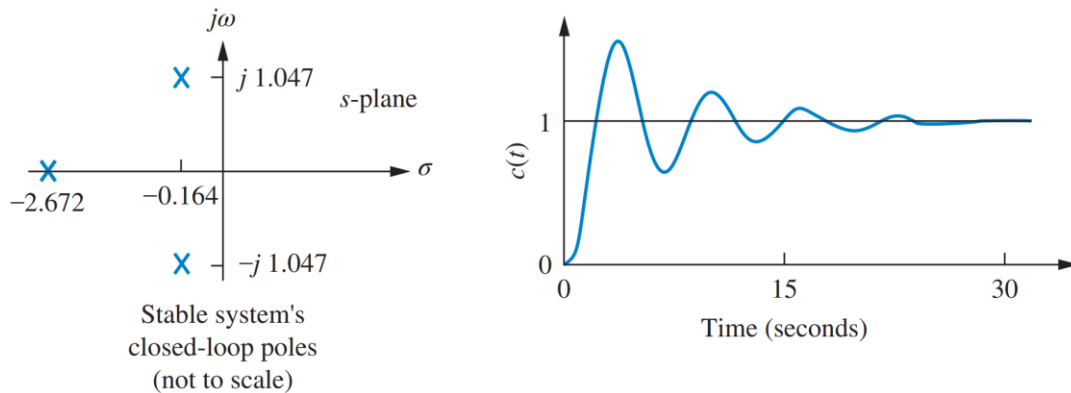
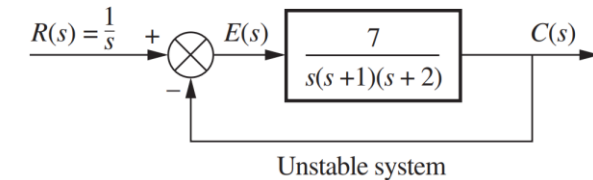
Bounded input bounded output (BIBO) stability



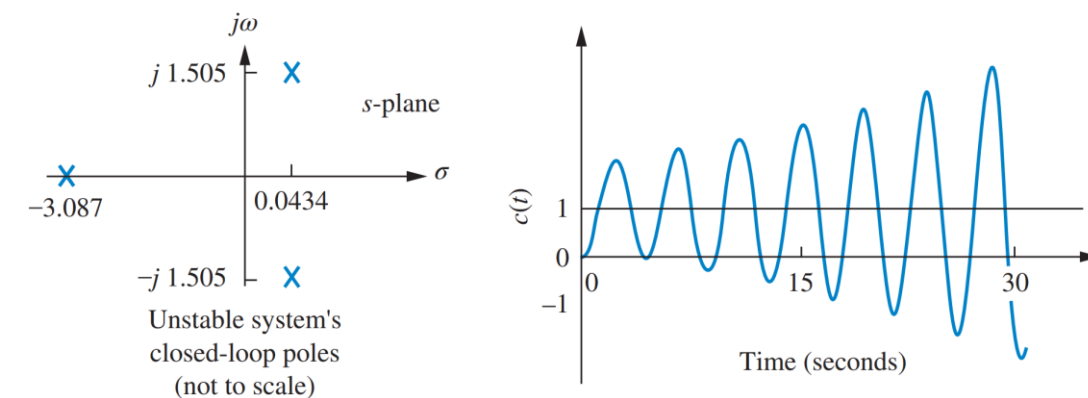
- ❑ A system is **stable** if every bounded input yields a bounded output.
- ❑ A system is **unstable** if any bounded input yields an unbounded output.
- ❑ **Marginal stability** means that the system is stable for some bounded inputs and unstable for others.
 - Poles of multiplicity greater than 1 on the imaginary axis lead to the sum of responses of the form $At^n \cos(\omega t + \phi)$, where $n = 1, 2, 3 \dots$ where the amplitude approaches infinity as time approaches infinity.



Increase gain from 3 to 7



(a)



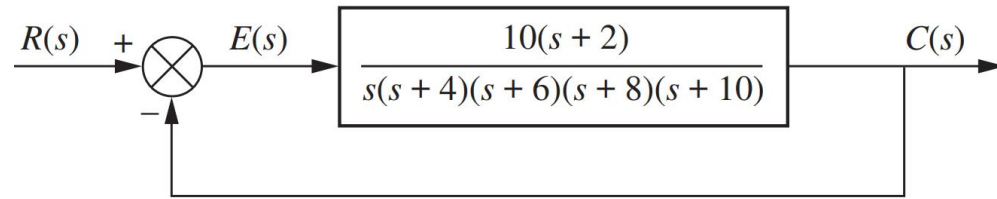
(b)

FIGURE 6.1 Closed-loop poles and response: **a.** stable system; **b.** unstable system

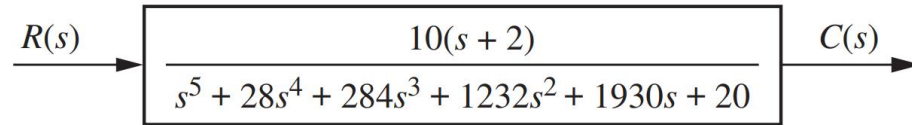


Stability

(in terms of Routh-Hurwitz Criterion)



(a)



(b)

FIGURE 6.2 Common cause of problems in finding closed-loop poles: **a.** original system; **b.** equivalent system

- We can quickly learn the number of poles in the right half s -plane and $j\omega$ -axis by writing down a table of coefficients, called Routh table.

TABLE 6.1 Initial layout for Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2			
s^1			
s^0			

Routh Table

Basic Routh table example

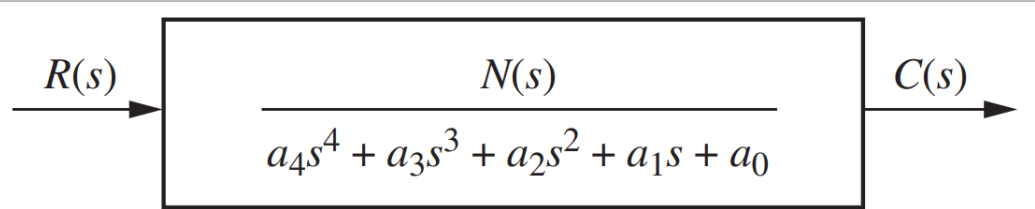


FIGURE 6.3 Equivalent closed-loop transfer function

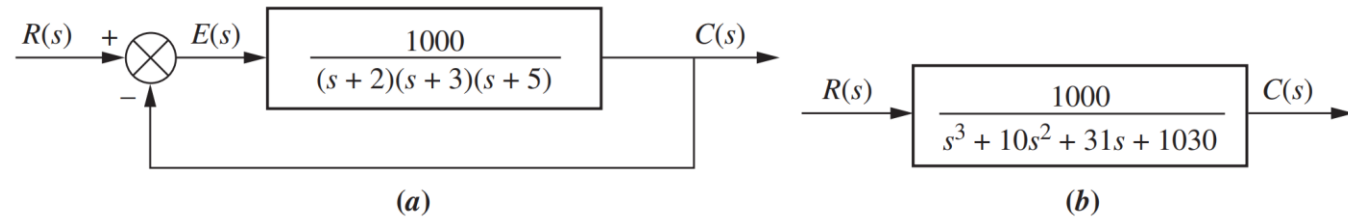
TABLE 6.2 Completed Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$-\frac{\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$-\frac{\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$-\frac{\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$-\frac{\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$-\frac{\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$-\frac{\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$-\frac{\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

Creating a Routh Table

PROBLEM: Make the Routh table for the system shown in Figure 6.4(a).

FIGURE 6.4 a. Feedback system for Example 6.1; b. equivalent closed-loop system



- Simply stated, the **Routh-Hurwitz criterion** declares that the number of roots of the polynomial that are in the right half-plane is equal to the number of sign changes in the first column of Routh table

PROBLEM: Make a Routh table and tell how many roots of the following polynomial are in the right half-plane and in the left half-plane.

$$P(s) = 3s^7 + 9s^6 + 6s^5 + 4s^4 + 7s^3 + 8s^2 + 2s + 6$$

ANSWER:

Routh Table with Zero Only in the First Column

- ❑ If the first element of a row is zero, division by zero would be required to form the next row. To avoid this phenomenon, an epsilon, ϵ , is assigned to replace the zero in the first column. The value ϵ is then allowed to approach zero from either the positive or the negative side, after which the signs of the entries in the first column can be determined.

---known as **epsilon method**.

TryIt 6.1

Use the following MATLAB statement to find the poles of the closed-loop transfer function in Eq. (6.1).

`roots([ 1 2 3 6 5 3])`

Stability via Epsilon Method

PROBLEM: Determine the stability of the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$$

(6.2)

SOLUTION: The solution is shown in Table 6.4. We form the Routh table by using the denominator of Eq. (6.2). Begin by assembling the Routh table down to the row where a zero appears *only* in the first column (the s^3 row). Next replace the zero by a small number, ϵ , and complete the table. To begin the interpretation, we must first assume a sign, positive or negative, for the quantity ϵ . Table 6.5 shows the first column of Table 6.4 along with the resulting signs for choices of ϵ positive and ϵ negative.

TABLE 6.4 Completed Routh table for Example 6.2

s^5	1	3	5
s^4	2	6	3
s^3	0 ϵ	$\frac{7}{2}$	0
s^2	$\frac{6\epsilon - 7}{\epsilon}$	3	0
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	0	0
s^0	3	0	0

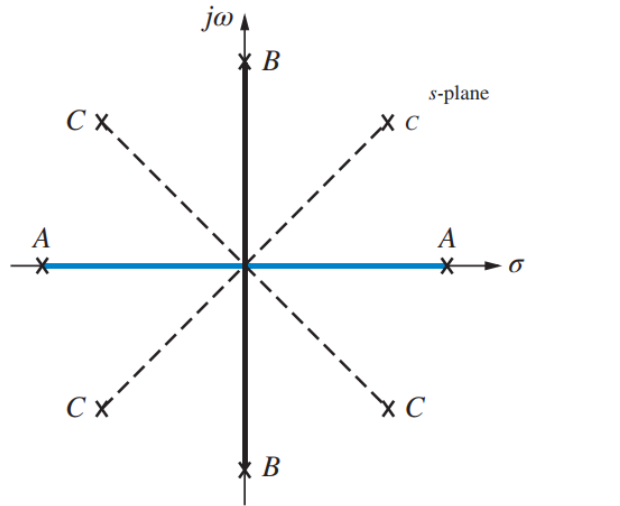
TABLE 6.5 Determining signs in first column of a Routh table with zero as first element in a row

Label	First column	$\epsilon = +$	$\epsilon = -$
s^5	1	+	+
s^4	2	+	+
s^3	0 ϵ	+	-
s^2	$\frac{6\epsilon - 7}{\epsilon}$	-	+
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	+	+
s^0	3	+	+

Routh Table with Entire Row being Zero

- When there is an even polynomial that is a factor of the original polynomial, an entire row consists of zeros will result in Routh table. We can write down an auxiliary polynomial and take derivative w.r.t. s to continue with the Routh table

---known as **auxiliary polynomial method**.



A: Real and symmetrical about the origin
 B: Imaginary and symmetrical about the origin
 C: Quadrantal and symmetrical about the origin

FIGURE 6.5 Root positions to generate even polynomials: A, B, C, or any combination

Stability via Routh Table with Row of Zeros

PROBLEM: Determine the number of right-half-plane poles in the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56} \quad (6.8)$$

SOLUTION: Start by forming the Routh table for the denominator of Eq. (6.8) (see Table 6.7). At the second row we multiply through by $1/7$ for convenience. We stop at the third row, since the entire row consists of zeros, and use the following procedure. First we return to the row immediately above the row of zeros and form an auxiliary polynomial, using the entries in that row as coefficients. The polynomial will start with the power of s in the label column and continue by skipping every other power of s . Thus, the polynomial formed for this example is

$$P(s) = s^4 + 6s^2 + 8 \quad (6.9)$$

Next we differentiate the polynomial with respect to s and obtain

$$\frac{dP(s)}{ds} = 4s^3 + 12s + 0 \quad (6.10)$$

Finally, we use the coefficients of Eq. (6.10) to replace the row of zeros. Again, for convenience, the third row is multiplied by $1/4$ after replacing the zeros.

The remainder of the table is formed in a straightforward manner by following the standard form shown in Table 6.2. Table 6.7 shows that all entries in the first column are positive. Hence, there are no right-half-plane poles.

TABLE 6.7 Routh table for Example 6.4

s^5	1	6	8
s^4	1	6	8
s^3	0	0	0
s^2	3	8	0
s^1	$\frac{1}{3}$	0	0
s^0	8	0	0



- 4.7 System Response with Additional
- 4.8 System Response with Zeros

Dominant Pole Concept



上海科技大学
ShanghaiTech University

Response of systems with zeros



上海科技大学
ShanghaiTech University